



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Employee Management System

The domain of the Project:
Salesforce

Team Mentors (and their designation):
Sriram Yarrbothula
Salesforce Developer

Team Members:

Mr.Aman Parmar
Mr. Nitish Kumar
Mr.Prateek Gupta
Mr.Kaushal Tiwari
Ms.Saloni Kukreti

Period of the project

March 2026 to April2026



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Declaration

The project titled “**Employee Management System**” has been mentored by **Sriram YarrBothula sir**, organised by SURE Trust, from **March 2026 to April 2026**, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. ----- -- Signature

Mr.Aman Parmar

Mr.Nitish Kumar

Mr.Prateek Gupta

Mr.Kaushal Tiwari

Ms.Saloni Kukreti

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Salesforce Developer – RAYON TECHNOLOGY SOLUTION PRIVATE LIMITED

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Executive Summary

- The Employee Management System is a cloud-based application developed
- using the Salesforce platform to manage and automate core Human Resource (HR) operations within an organization.
- The system focuses on handling employee information, attendance tracking, leave management, and payroll processing in a centralized and efficient manner.
- The primary aim of this project is to replace traditional manual methods with a digital solution
- that ensures accuracy, reduces redundancy, and improves overall productivity.
- By using Salesforce features such as custom objects, validation rules, and Lightning Web Components (LWC),
- the system provides a structured and user-friendly interface for managing employee data.
- In addition, the project includes reports and dashboards that offer real-time insights into employee activities, attendance trends, and payroll summaries
- This helps management in better decision-making.
- Overall, the system is scalable, reliable, and aligned with modern industry practices.



Introduction

- Background and context of the project
- **Problem statement or goals of the project**
- Scope and limitations of the project
- Innovation component in the project
- **The Employee Management System** is a Salesforce-based application designed to efficiently manage employee details, departments, attendance, leave requests, and payroll processes. It helps automate HR operations, improve data organization, and enhance overall productivity using Salesforce tools like Apex, Flows, and Lightning Web Components.

Background and context of the project : The Employee Management System is a Salesforce-based application designed to efficiently manage employee details, departments, attendance, leave requests, and payroll processes. It helps automate HR operations, improve data organization, and enhance overall productivity using Salesforce tools like Apex, Flows, and Lightning Web Components.

- **Problem statement or goals of the project :**
 - 1 :Manual handling of HR processes often results in
 - 2: Data inconsistency Increased chances of errors
 - 3: Difficulty in tracking employee activities
 - 4:Time-consuming operations
- **Scope and limitations of the project:** The goal of this project is to develop a centralized Employee Management System using Salesforce that automates these processes and ensures smooth data management.
- Scope of the
 - 1.Project The system covers:
 - 2.Employee data management
 - 3.Attendance tracking system Leave request and approval process
 - 4.Payroll management
 5. Reporting and dashboard
 - 6 visualization

Limitations : Payroll module includes only basic salary structure No integration with external hardware/software (e.g., biometric systems) Limited advanced analytics

Innovation in the Project : The use of Salesforce cloud technology along with Lightning Web Components provides: A modern and scalable solution Real-time data access Interactive and user-friendly interface



Project Objectives

- Clearly defined objectives and goals of the project
- Expected outcomes and deliverables.
- **Project Objective:** The main objectives of this project are:
 1. To design and develop a centralized system for employee data management
 2. To automate HR processes such as attendance, leave, and payroll
 3. To reduce manual effort and minimize human errors
 4. To ensure data accuracy using validation rules
 5. To provide a simple and efficient user interface
 6. To enable real-time tracking of employee activities
 7. To generate useful reports and dashboards
 8. To improve overall efficiency and productivity of the organization
 9. To create a scalable system that can be enhanced in the future

Methodology and Results : The development of this project was carried out in a structured manner:

Requirement Analysis Understanding the needs of an HR system, including employee management, attendance tracking, leave handling, and payroll processing.

2. System Design Created custom objects such as Employee, Department, Attendance, Leave, and Payroll Defined relationships between objects to maintain data integrity

3. Implementation

Added fields to store relevant data (name, ID, salary, dates, etc.)

Applied validation rules to prevent incorrect data entry

Developed Lightning Web Components (LWC) for interactive UI Configured approval processes for leave management

1.Reporting and Dashboard Creation Built reports to analyze employee and attendance data
Designed dashboards for visual representation of key metrics

5. Testing Tested all modules to ensure proper functionality Verified validation rules and data flow

6. Deployment Successfully deployed the system within the Salesforce environment



Methodology and Results

- Methods/Technology used
- Tools/Software used
- Data collection approach (if applicable)
- Project Architecture (Clearly describe the architecture)
- Final project working screenshots along with supporting explanation
- Project GitHub Link

- **Methods/Technology used**

1 : **Data Modeling** : Custom objects were created to store and manage data

Employee__c – Stores employee details

Department__c – Stores department information

Attendance__c – Tracks employee attendance

Leave_Request__c – Manages leave applications

Payroll__c – Handles salary processing

The screenshot displays the Salesforce Object Manager interface for the 'Employees' object. The left sidebar shows navigation options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main area is titled 'Fields & Relationships' and shows a list of 14 items, sorted by Field Label. The table below represents the data shown in the screenshot:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Department	Departmentt__c	Lookup(Departmentt)		✓
Email	Emails__c	Email (External ID)		✓
Employee ID	Employee_ID__c	Auto Number		
Employees Name	Name	Text(80)		

At the bottom right of the screenshot, there is a watermark that reads 'Activate Windows Go to Settings to activate Windows'.



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The screenshot shows the Salesforce Setup interface for the 'Employees' object. The left sidebar contains a list of setup options: Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, Triggers, Flow Triggers, Validation Rules, and Conditional Field Formatting. The main area is titled 'Fields & Relationships' and displays a table of 14 items, sorted by Field Label. The table includes fields like Owner, Phone, Record Type, Role, Salary, and Status, each with its corresponding field ID and data type. A 'Quick Find' search bar is located at the top right of the table. The interface also includes a 'Developer Edition' badge and a 'Search Setup' bar.

Field Label	Field ID	Data Type	Relationship
Owner	OwnerId	Lookup(User,Group)	✓
Phone	Phone__c	Phone	
Record Type	RecordTypeId	Record Type	✓
Role	Role__c	Picklist	
Salary	Salary__c	Currency(18, 0)	
Status	Status__c	Picklist	

2. Relationships

Department → Employee (One-to-Many)

Employee → Attendance, Leave, Payroll

The screenshot shows the Salesforce Setup interface for the 'Employees' object, specifically the 'Details' section. The left sidebar contains a list of setup options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main area displays the details for the 'Department__c' field, including its API Name, Description, Help Text, Data Owner, Field Usage, Data Sensitivity Level, and Compliance Categorization. It also shows the 'Lookup Options' section, which includes a 'Related To' dropdown set to 'Department__c', a 'Child Relationship Name' of 'Employees', and a 'Required' checkbox. The 'Lookup Filter' section indicates that no lookup filters are defined. The interface also includes a 'Developer Edition' badge and a 'Search Setup' bar.

Field Label	Field ID	Data Type	Relationship
Department	Department__c	Lookup	



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The screenshot shows the Salesforce Setup interface in Developer Edition. The breadcrumb trail is **SETUP > OBJECT MANAGER**. The main heading is **Leave Request**. On the left sidebar, the navigation menu includes: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area displays **Validation Rules** for the 'Leave Request' object, showing 1 item sorted by Rule Name. A table lists the validation rule:

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Leave_Request_Check	From Date	From Date cannot be greater than To Date	✓	Salesforce Project, 28/04/2026, 8:49 pm

A 'New' button is located in the top right corner of the Validation Rules section. An 'Activate Windows' watermark is visible at the bottom right.

3.Validation Rules

- Salary must be greater than zero
- Leave dates must be valid
- Email must follow proper format
- Attendance cannot be marked for future dates

The screenshot shows the Salesforce Setup interface in Developer Edition. The breadcrumb trail is **SETUP > OBJECT MANAGER**. The main heading is **Employees**. On the left sidebar, the navigation menu includes: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area displays **Validation Rules** for the 'Employees' object, showing 2 items sorted by Rule Name. A table lists the validation rules:

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Email_required	Top of Page	Email is required	✓	Salesforce Project, 27/03/2026, 3:57 pm
Salary_Check	Top of Page	Salary must be greater than 0	✓	Salesforce Project, 27/03/2026, 3:50 pm

A 'New' button is located in the top right corner of the Validation Rules section. An 'Activate Windows' watermark is visible at the bottom right.

4. Security Implementation

- Organization-Wide Defaults set to Private
- Role hierarchy: HR > Manager > Employee
- Profiles created for different roles

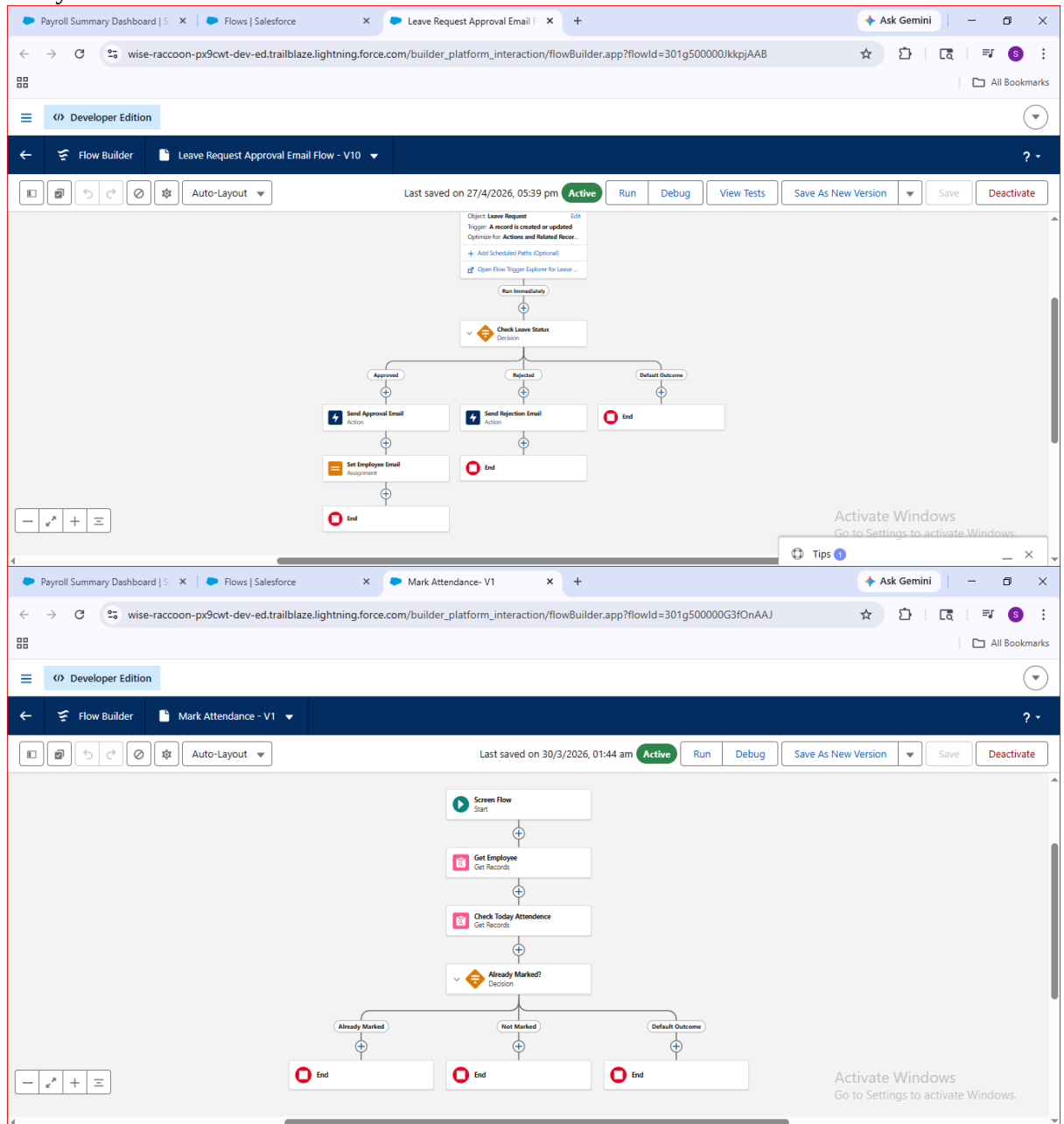


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5. Automation

• Flows:

- Leave request automatically set to “Pending”
- Email notifications on approval
- Daily attendance check



• Apex:

- Prevent duplicate attendance entries
- Monthly payroll generation using batch Apex



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```
Developer Console - Google Chrome
wise-raccoon-px9cwt-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >
MonthlyPayrollBatch.apxc PreventDuplicateAttendance.apxt
Code Coverage: None API Version: 66 Go To

1 trigger PreventDuplicateAttendance on Attendance__c (before insert, before update) {
2
3     Set<Id> employeeIds = new Set<Id>();
4     Set<Date> dates = new Set<Date>();
5
6     for (Attendance__c att : Trigger.new) {
7         if (att.Employees__c != null && att.Date__c != null) {
8             employeeIds.add(att.Employees__c);
9             dates.add(att.Date__c);
10        }
11    }
12
13    List<Attendance__c> existingRecords = [
14        SELECT Id, Employees__c, Date__c
15        FROM Attendance__c
16        WHERE Employees__c IN :employeeIds
17        AND Date__c IN :dates
18    ];
19
20    for (Attendance__c newRec : Trigger.new) {
21        for (Attendance__c existing : existingRecords) {
22
23            if (newRec.Employees__c == existing.Employees__c &&
```

Activate Windows
Go to Settings to activate Windows.

Logs, Tests, and Problems

```
Developer Console - Google Chrome
wise-raccoon-px9cwt-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >
MonthlyPayrollBatch.apxc PreventDuplicateAttendance.apxt
Code Coverage: None API Version: 66 Go To

9         dates.add(att.Date__c);
10    }
11}
12
13List<Attendance__c> existingRecords = [
14    SELECT Id, Employees__c, Date__c
15    FROM Attendance__c
16    WHERE Employees__c IN :employeeIds
17    AND Date__c IN :dates
18];
19
20for (Attendance__c newRec : Trigger.new) {
21    for (Attendance__c existing : existingRecords) {
22
23        if (newRec.Employees__c == existing.Employees__c &&
24            newRec.Date__c == existing.Date__c &&
25            newRec.Id != existing.Id) {
26
27            newRec.addError('Attendance already marked for this employee on this date');
28        }
29    }
30}
31}
```

Activate Windows
Go to Settings to activate Windows.

Logs, Tests, and Problems



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```
Developer Console - Google Chrome
wise-raccoon-px9cwt-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >
MonthlyPayrollBatch.apxc PreventDuplicateAttendance.apxt
Code Coverage: None API Version: 66 Go To

1 global class MonthlyPayrollBatch implements Database.Batchable<SObject> {
2
3     global Database.QueryLocator start(Database.BatchableContext bc) {
4         return Database.getQueryLocator(
5             [SELECT Id, Name, Salary__c, Status__c FROM Employees__c WHERE Status__c = 'Active']
6         );
7     }
8
9     global void execute(Database.BatchableContext bc, List<SObject> scope) {
10
11         List<Payroll__c> payrollList = new List<Payroll__c>();
12
13         String currentMonth = String.valueOf(Date.today().month()) + '-' + String.valueOf(Date.today().year());
14         system.debug(currentMonth);
15
16         for(SObject obj : scope){
17
18             Employees__c emp = (Employees__c) obj;
19
20             Decimal baseSalary = emp.Salary__c;
21             Decimal deductions = baseSalary * 0.1;
22
23             Payroll__c payroll = new Payroll__c(

```

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Go to Settings to activate Windows.

```

19
20         Decimal baseSalary = emp.Salary__c;
21         Decimal deductions = baseSalary * 0.1;
22
23         Payroll__c payroll = new Payroll__c(
24             Employees__c = emp.Id, // ✓ FIXED HERE
25             Month__c = currentMonth,
26             Base_Salary__c = baseSalary,
27             Deductions__c = deductions
28         );
29
30         payrollList.add(payroll);
31     }
32
33     if(!payrollList.isEmpty()){
34         insert payrollList;
35     }
36 }
37
38 global void finish(Database.BatchableContext bc){
39     System.debug('Payroll Batch Completed');
40 }
41 }

```

Activate Windows
Go to Settings to activate Windows.

Logs, Tests, and Problems

6.Lightning Web Components

- Employee Dashboard



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The screenshot displays the Lightning Studio interface for developing a Salesforce component. The top panel shows the HTML template for the `employeeDashboard.html` component, which includes conditional rendering for employee profile and attendance information. The bottom panel shows the JavaScript controller logic for fetching employee data. The logs panel at the bottom shows successful resource retrieval and data fetching.

HTML Template (employeeDashboard.html):

```
<template if:true={employee}>
  <!-- PROFILE CARD -->
  <div class="card profile">
    <h2>{{employee.Name}}</h2>
    <p>{{employee.Emails__c}}</p>

    <template if:true={employee.Department__r}>
      <p>{{employee.Department__r.Name}}</p>
    </template>
  </div>

  <!-- ATTENDANCE -->
  <div class="attendance-container">
    <div class="card present">
      <h3>{{presentCount}} Present</h3>
      <p>{{presentCount}}</p>
    </div>

    <div class="card absent">
      <h3>{{absentCount}} Absent</h3>
      <p>{{absentCount}}</p>
    </div>
  </div>
</template>

<template if:false={employee}>
  <p class="loading">Loading Dashboard...</p>
</template>
```

JavaScript Controller (employeeDashboard.js):

```
1 import { LightningElement, wire } from 'lwc';
2 import getEmployeeData from '@salesforce/apex/EmployeeController.getEmployeeData';
3
4 export default class EmployeeDashboard extends LightningElement {
5   employee;
6   presentCount = 0;
7   absentCount = 0;
8
9   @wire(getEmployeeData)
10  wiredData({ data, error }) {
11    if (data) {
12      this.employee = data.employee;
13      this.presentCount = data.present;
14      this.absentCount = data.absent;
15    } else if (error) {
16      console.error(error);
17    }
18  }
19 }
```

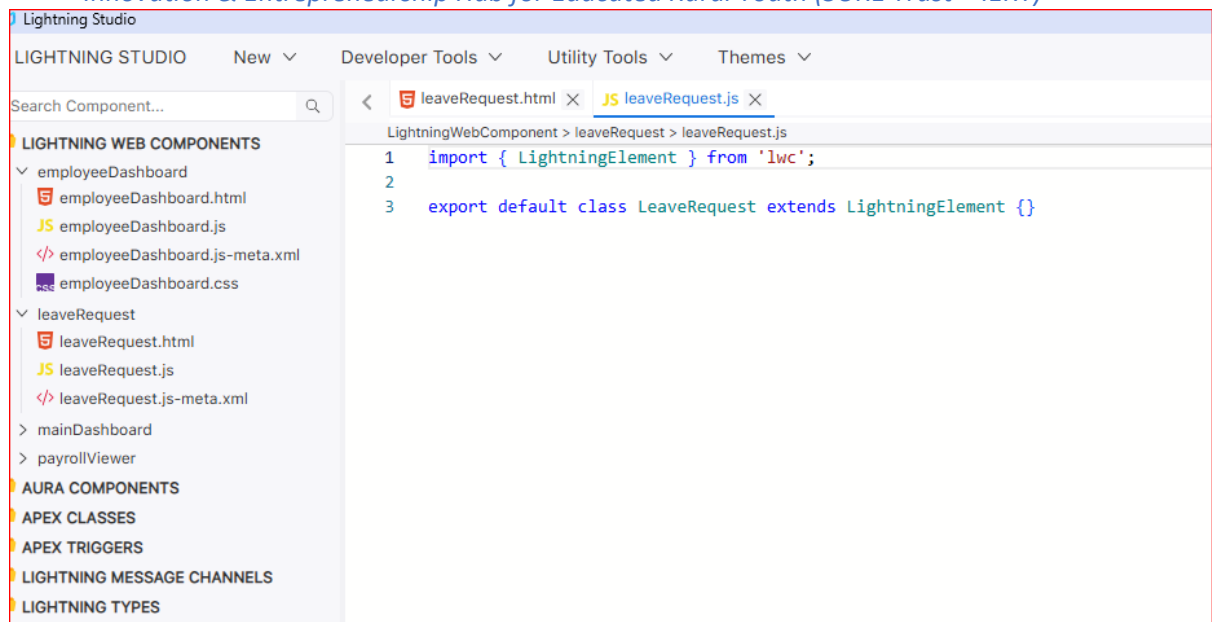
Logs:

System	Message	Time
SYSTEM	Retrieving resources, Please wait...	22:11:08
SUCCESS	Successfully retrieved resources...	22:11:10
FETCH	employeeDashboard • LightningWebComponent • Fetch process started. Please wait...	22:11:16
SUCCESS	employeeDashboard • Fetched successfully.	22:11:16
FETCH	employeeDashboard • LightningWebComponent • Fetch process started. Please wait...	22:11:22
SUCCESS	employeeDashboard • Fetched successfully.	22:11:23

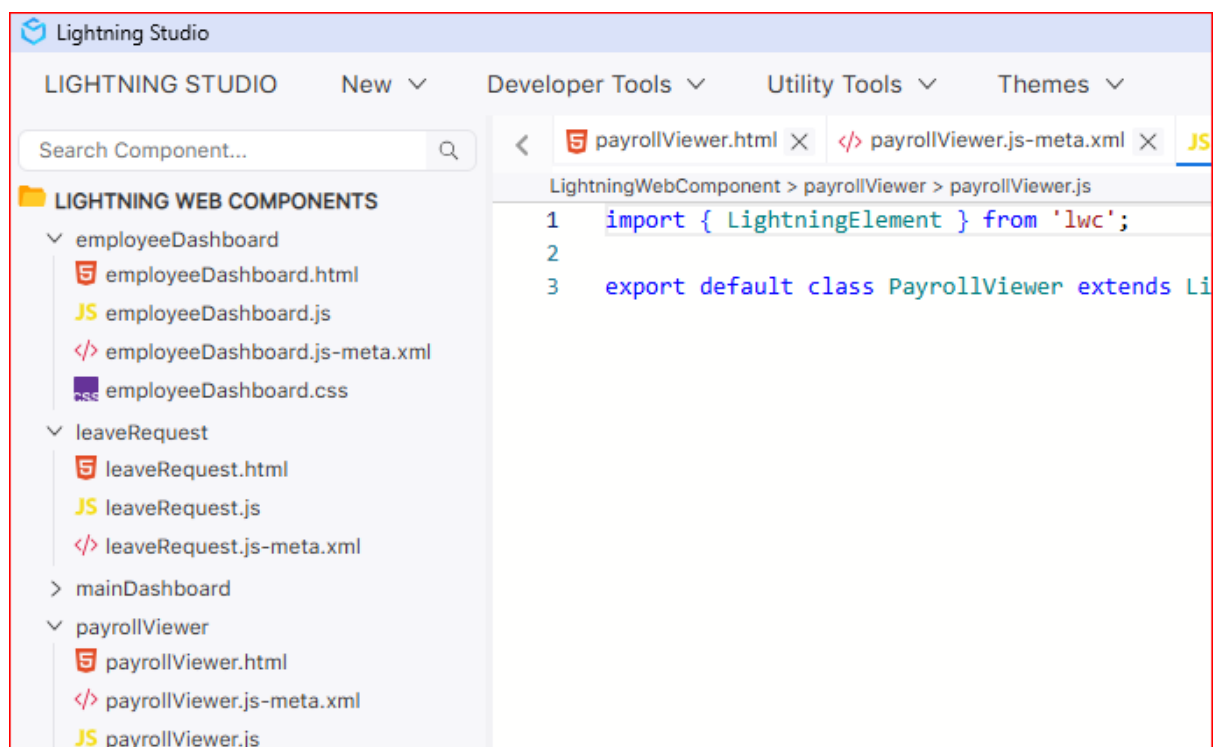
- Leave Request Component



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- Payroll Viewer

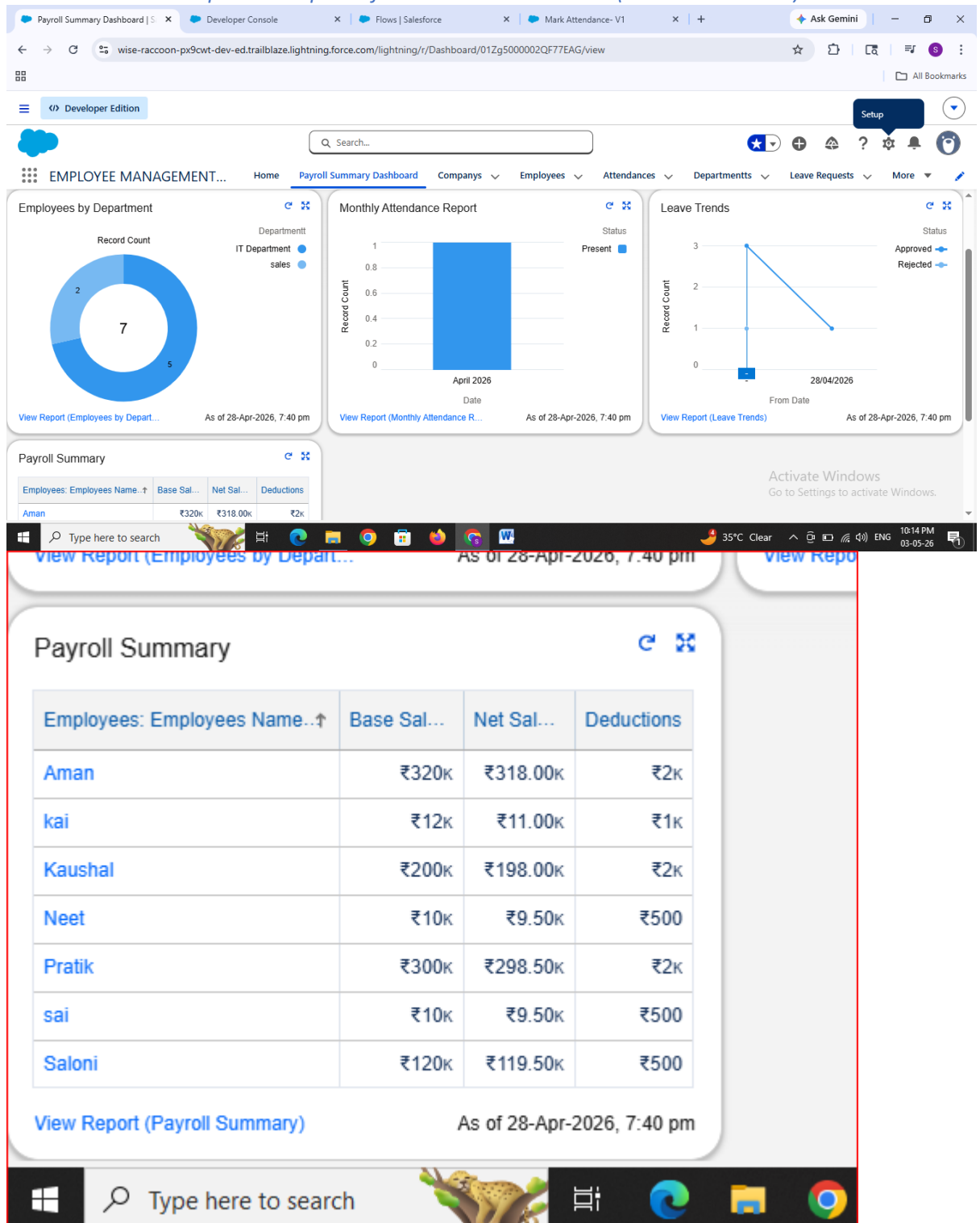


7. Reports & Dashboards

- Employee distribution by department
- Attendance reports
- Leave trends
- Payroll summary



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Results

- Successfully developed a fully functional Employee Management System
- Reduced manual effort through automation
- Improved data accuracy and consistency



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- Enabled real-time monitoring using dashboards
- Created a scalable solution suitable for organizations



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Learning and Reflection

- Document every team member's new learnings (in terms of technology, management, etc)
- Document every team member's experience. (overall experience

LEARNING & REFLECTION

Technical Learning

- Salesforce data modeling and object relationships
- Automation using Flow Builder
- Apex programming concepts
- Lightning Web Component development
- Security and access control implementation

Personal Learning

- Problem-solving and analytical thinking
- Understanding real-world business requirements
- Time management and project planning
- Hands-on experience with industry tools



Conclusion and Future Scope

- Recap objectives and achievements
- Future scope of this project

Conclusion

The Employee Management System project successfully demonstrates how Salesforce can be used to build a complete HR management solution. It integrates multiple functionalities such as employee management, attendance tracking, leave handling, and payroll processing into a single platform.

- The Employee Management System was successfully developed using the Salesforce platform.
 - The project effectively manages employee data, attendance, leave, and payroll in a centralized system.
 - Automation using Flows and Apex reduced manual work and improved efficiency.
 - Validation rules ensured data accuracy and consistency.
 - Role-based security provided controlled access to sensitive information.
 - The system improved productivity and simplified HR operations.
 - Real-time reports and dashboards helped in better decision-making.
 - The project demonstrates practical implementation of Salesforce in real-world business scenarios.
 - Overall, the system is scalable, reliable, and suitable for organizationa
- **Future scope of this project:**
 - Integration with external HR or payroll systems
 - Mobile application development for easy access
 - AI-based analytics for employee performance
 - Advanced reporting and predictive insights